Breadth-first search

For an unweighted graph, breadth-first search explores vertices by distance in number of edges from its start. It uses a FIFO queue and can find minimum-edge paths. Mark a vertex visited when enqueuing it to prevent duplicate enqueues.

Depth-first search

Depth-first search follows a branch as far as possible before backtracking. Its pending path can be represented by the recursive call stack or an explicit stack. Mark each vertex visited before exploring its unvisited neighbors to avoid revisiting cycles.

Undirected degrees

For a finite undirected graph, the sum of all vertex degrees equals twice the number of edges. Each edge contributes one incidence at each endpoint; a self-loop contributes two. Thus a graph with 10 edges has degree sum 20.

Adjacency matrix

An adjacency matrix for V vertices has V rows and V columns and uses O(V squared) storage. It checks whether a specified edge exists by reading one cell in O(1) time. This can waste space for a sparse graph compared with adjacency lists.

Dijkstra edge requirement

Dijkstra's shortest-path algorithm assumes nonnegative edge weights. It repeatedly chooses the smallest tentative distance using a min-priority queue. Negative-weight edges invalidate its usual greedy correctness guarantee.